

WTW 158 Worksheet 3 on functions, inverses, exponentials and logarithms

A. STUDY GUIDE PROBLEMS

p 33 no 2, 3, 4 p42 no 3, 17, 23, 35 p53 no 17, 19
p66 no 21, 31 p 66 no 23, 38, 41, 51, 56

B. MORE PROBLEMS

1. Sketch the function and determine the domain and range of the function.

a. $m(x) = \ln(1+x)$ b. $n(x) = 1 - \ln x$ c. $h(x) = \ln|x|$ d. $l(x) = |\ln x|$

2. Sketch the function and determine the domain and range of the function.

a. $m(x) = e^{1+x}$ b. $n(x) = 1 - e^{-x}$ c. $h(x) = e^{|x|}$ d. $l(x) = |e^x|$

3. Solve for x :

(i) $\ln(2x^2 + x) \leq 0$ (ii) $|\ln x| \leq 1$

(iii) $\ln|x| \leq 0$ (Use a graph) (iv) $e^x(x-1) > 0$

(v) $e^{x-1} \ln x > 0$ (vi) $e^x \ln|x| > 0$

4. If $(x) = \frac{1}{x-1}$ and $g(x) = \ln x$, find $f \circ g(x)$ and determine the domain of $f \circ g$.

5. For $f(x) = e^{1+x}$ (2a. above), find the inverse function f^{-1} and sketch both f and f^{-1} .

C. TEST QUESTIONS

Question 1 : If $2\ln x^3 - 4\ln x + \ln 5x = \ln a$, then $a =$

[a] $2x^3 + x$	[b] $5x^3$	[c] $\frac{x}{5}$	[d] None of these
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Question 2 : The solution set of $\ln(x^2) - 6 > 0$ is

[a] (e^3, ∞)	[b] $(-\infty, e^3)$	[c] $(-\infty, -e^3)$	[d] $(-\infty, -e^3) \cup (e^3, \infty)$
[e] $(-e^3, e^3)$	[f] None of these		

Question 3: If $xe^{-x} + 2e^{-x} = 0$, then

(a) $x = -2$	(b) $x = -2$ or / of $x = 0$	(c) $x = 0$ or / of $x = 1$
(d) $x = 0$	(e) $x = -2$ or / of $x = 1$	(f) None of these

Question 4: The domain of the function $f(x) = \frac{1}{1 + \ln(x+1)}$ is

(a) $\left(-1 + \frac{1}{e}, \infty\right)$	(b) $(-1, \infty)$	(c) $\left(-1, -1 + \frac{1}{e}\right) \cup \left(-1 + \frac{1}{e}, \infty\right)$
(d) $\left(-1, -1 + \frac{1}{e}\right)$	(e) $x \in \mathbb{R}, x \neq -1 + \frac{1}{e}$	(f) None of these

Question 5: Find the domain of $f(x) = \frac{\ln(x+1)}{(x-1)^2}$.

D YOU TUBE videos: Natural base (e) and Natural logarithm (ln) Kahn Academy
Properties of logarithms Part 1 and Part 2 Patrickjmt

SELECTED ANSWERS:

B. 3 (i) $\left[-1, -\frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right]$ (ii) $\left[\frac{1}{e}, e\right]$ (iii) $[-1, 1] \setminus \{0\}$ (iv) $(1, \infty)$ (v) $(1, \infty)$ (vi) $(-\infty, -1) \cup (1, \infty)$

4. $f \circ g(x) = \frac{1}{\ln x - 1}$ Domain: $(0, \infty) \setminus \{e\}$ 5. $y = \ln x - 1$

C. Q1. [b] Q2. [d] Q3 (a) Q4 (e) Q5 $[-1, \infty) \setminus \{1\}$